



On-Hand Solution of the Problem (Harvey's Specialty Shop)

In this section, to prove the accuracy of the algorithm that will be used during software development, the (Q,R) inventory model parameters for the given test problem (Harvey's Specialty Shop) have been solved step-by-step using the iteration method. To enable the software development team to build the algorithm easily, the inputs and mathematical relationships are defined first.

1. Initial Parameters (Inputs)

The data obtained from the test problem are the constant inputs of the system:

- **c (Unit Cost):** \$20
- **K (Ordering Cost):** \$100
- **p (Penalty/Shortage Cost):** \$20
- **I (Annual Interest/Holding Rate):** 0.25 (25%)
- **LT (Lead Time):** 4 months
- **μ (Average Demand During Lead Time):** 500 jars
- **σ (Standard Deviation of Lead Time Demand):** 100

Pre-calculations:

The algorithm must calculate the following two values before entering the loop:

- **Annual Demand (λ):** Since the lead time is 4 months and the average demand during this period is 500, the 12-month (annual) demand is calculated as: $\lambda = 500 \cdot \left(\frac{12}{4}\right) = 1500$.
- **Holding Cost (h):** $h = I \cdot c = 0.25 \cdot 20 = 5$.

2. Algorithm Loop Formulas

The computer program will run the following formulas sequentially within a loop (iteration). The loop must continue until the newly found Q and R values remain the same as the previous iteration's values.

1. Order Quantity (**Q**):

$$Q = \sqrt{\left(\frac{2\lambda[K + p \cdot n(R)]}{h}\right)}$$

2. Probability of No Stockout (**F(R)**):

$$1 - F(R) = \frac{Q \cdot h}{p \cdot \lambda}$$

3. Reorder Point (**R**) Calculation: The calculated F(R) value must be looked up in the Z-Table to find the corresponding z value. Then:

$$R = \mu + z \cdot \sigma$$

4. Expected Shortage per Cycle (**n(R)**): The found z value must be looked up in the Loss Function Table to find L(z). Then:

$$n(R) = \sigma \cdot L(z)$$

3. Step-by-Step Iterations

Initialization (Iteration 0): To start the algorithm, it is initially assumed that there is no shortage ($n(R) = 0$).

- $Q_0 = \sqrt{\frac{2 \cdot 1500 \cdot 100}{5}} \approx 245$
- $1 - F(R_0) = \frac{245 \cdot 5}{20 \cdot 1500} \approx 0.0408$
- $F(R_0) = 0.9592$
- Corresponding value from Z-Table: $Z_0 = 1.74$
- Corresponding value from Loss Function Table: $L(1.74) = 0.0166$
- $n(R_0) = 100 \cdot 0.0166 = 1.66$

Iteration 1: The loop begins using the $n(R_0) = 1.66$ value found in the previous step.

- $Q_1 = \sqrt{\frac{2 \cdot 1500 \cdot [100 + (20 \cdot 1.66)]}{5}} \approx 283$
- $1 - F(R_1) = \frac{283 \cdot 5}{20 \cdot 1500} \approx 0.0471$
- $F(R_1) = 0.9528$
- Z-Table lookup: $Z_1 = 1.67$
- Loss Function lookup: $L(1.67) = 0.0196$
- $R_1 = 500 + 1.67 \cdot 100 = 667$
- $n(R_1) = 100 \cdot 0.0196 = 1.96$

Iteration 2: The $n(R_1) = 1.96$ value from the previous step is used.

- $Q_2 = \sqrt{\frac{2 \cdot 1500 \cdot [100 + (20 \cdot 1.96)]}{5}} \approx 289$
- $1 - F(R_2) = \frac{289 \cdot 5}{20 \cdot 1500} = 0.0482$
- $F(R_2) = 0.9518$
- Z-Table lookup: $Z_2 = 1.66$
- Loss Function lookup: $L(1.66) = 0.0201$
- $R_2 = 500 + 1.66 \cdot 100 = 666$
- $n(R_2) = 100 \cdot 0.0201 = 2.01$

Iteration 3 (Equilibrium State): The $n(R_2) = 2.01$ value from the previous step is used.

- $Q_3 = \sqrt{\frac{2 \cdot 1500 \cdot [100 + (20 \cdot 2.01)]}{5}} \approx 291$
- $1 - F(R_3) = \frac{291 \cdot 5}{20 \cdot 1500} = 0.0485$
- $F(R_3) = 0.9515$
- Z-Table lookup: $Z_3 = 1.66$ (Reached the same z value as the previous iteration)
- Loss Function lookup: $L(1.66) = 0.0201$
- $R_3 = 500 + 1.66 \cdot 100 = 666$
- $n(R_3) = 100 \cdot 0.0201 = 2.01$

Conclusion and Stopping Criteria: At the end of Iteration 3, the z value (1.66) and consequently the calculated $n(R)$ value have stabilized. Optimization is achieved, and the loop terminates.

- **Optimal Order Quantity (Q):** $Q = 291$
- **Reorder Point (R):** $R = 500 + 1.66 \cdot 100 = 666$
- **Number of Iterations:** The algorithm stopped at 3rd iteration (When z value stabilized at 1.66).
- **Safety Stock:** $SS = R - \mu = 666 - 500 = 166$
- **Average Annual Costs:**
 - **Annual Holding Cost:** $h \cdot \left(\frac{Q}{2} + SS\right) = 5 \cdot \left(\frac{291}{2} + 166\right) = \1557.50
 - **Annual Setup Cost:** $K \cdot \frac{\lambda}{Q} = 100 \cdot \frac{1500}{291} \approx \515.46
 - **Annual Penalty Cost:** $p \cdot \frac{\lambda}{Q} \cdot n(R) = 20 \cdot \frac{1500}{291} \cdot 2.01 \approx \207.22
- **Average Time Between Orders:** $T = \frac{Q}{\lambda} = \frac{291}{1500} = 0.194 \text{ years} (0.194 \cdot 12 = 2.328 \text{ months})$
- **Proportion of Order Cycles in Which No Stock Out Occurs:** $F(R_3) = 0.9515 = (95.15\%)$
- **Proportion of Demands That Are Not Met:** $\frac{n(R)}{Q} = \frac{2.01}{291} \approx 0.0069 = (0.69\%)$

Pseudocode of the Project:

```
// 1. INPUT PHASE
PRINT "Please enter the unit cost:"
INPUT unit_cost
PRINT "Please enter the ordering cost:"
INPUT ordering_cost
PRINT "Please enter the penalty cost:"
INPUT penalty_cost
PRINT "Please enter the annual interest rate as decimal:"
INPUT interest_rate

PRINT "Please enter the lead time in months:"
INPUT lead_time
PRINT "Please enter the average demand during lead time:"
INPUT mean_demand
PRINT "Please enter the standard deviation of lead time demand:"
INPUT std_dev

// 2. PRE-CALCULATIONS
SET annual_demand = mean_demand * (12 / lead_time)
SET holding_cost = interest_rate * unit_cost

// 3. INITIALIZATION FOR THE LOOP
SET expected_shortage = 0
SET previous_z_score = -1
SET current_z_score = 0
SET iteration_count = 0
SET max_iterations = 1000

// 4. ALGORITHM LOOP
WHILE (previous_z_score is NOT EQUAL TO current_z_score) AND (iteration_count < max_iterations) DO

    SET previous_z_score = current_z_score
    SET iteration_count = iteration_count + 1
```

```

// Calculate Order Quantity
SET order_quantity = SQUARE_ROOT( (2 * annual_demand * (ordering_cost +
(penalty_cost * expected_shortage))) / holding_cost )

// Calculate Probabilities
SET stockout_probability = (holding_cost * order_quantity) / (penalty_cost *
annual_demand)
SET no_stockout_probability = 1 - stockout_probability

// Lookup Values from Tables
SET current_z_score = LOOKUP_Z_TABLE(no_stockout_probability)
SET loss_function_value = LOOKUP_LOSS_FUNCTION(current_z_score)

// Update Expected Shortage
SET expected_shortage = std_dev * loss_function_value

END WHILE

// 5. FINAL PERFORMANCE MEASURE CALCULATIONS
SET final_order_quantity = ROUND(order_quantity)

// Calculate and round Reorder Point
SET exact_reorder_point = mean_demand + (current_z_score * std_dev)
SET reorder_point = ROUND(exact_reorder_point)

// Calculate Safety Stock
SET safety_stock = reorder_point - mean_demand

// Calculate Annual Costs
SET annual_setup_cost = ordering_cost * (annual_demand / final_order_quantity)
SET annual_holding_cost = holding_cost * ((final_order_quantity / 2) + safety_stock)
SET annual_penalty_cost = penalty_cost * (annual_demand / final_order_quantity) *
expected_shortage

// Calculate Time Between Orders
SET time_between_orders_years = final_order_quantity / annual_demand
SET time_between_orders_months = time_between_orders_years * 12

// Calculate Proportions
SET proportion_unmet_demand = expected_shortage / final_order_quantity

// 6. OUTPUT RESULTS
PRINT "Optimal Lot Size: " + final_order_quantity
PRINT "Reorder Point: " + reorder_point
PRINT "Number of Iterations: " + iteration_count
PRINT "Safety Stock: " + safety_stock
PRINT "Average Annual Holding Cost: " + annual_holding_cost
PRINT "Average Annual Setup Cost: " + annual_setup_cost
PRINT "Average Annual Penalty Cost: " + annual_penalty_cost
PRINT "Average Time Between Orders in months: " + time_between_orders_months
PRINT "Proportion of cycles with no stockout: " + no_stockout_probability
PRINT "Proportion of demands not met: " + proportion_unmet_demand

```

Sustainable Development Goals:

This project aligns with the United Nations Sustainable Development Goals by leveraging algorithmic optimization to improve resource management and industrial efficiency. The developed (Q,R) inventory software specifically contributes to the following goals:

Goal 9: Industry, Innovation, and Infrastructure

- **Innovation in Supply Chain:** The project introduces a digital solution to replace manual inventory tracking, fostering innovation in small-scale business operations.
- **Infrastructure Reliability:** By automating the Z-Table and Loss Function lookups, the software minimizes human error, ensuring a more resilient and reliable industrial infrastructure.
- **Data-Driven Decision Making:** The software provides precise decision variables (Q and R), allowing businesses to build a more robust operational framework based on statistical demand data.

Goal 12: Responsible Consumption and Production

- **Reduction of Overstocking:** By calculating the optimal lot size (Q), the software prevents unnecessary accumulation of inventory. This directly reduces waste, especially for products with a shelf life or potential for decay.
- **Resource Efficiency:** The algorithm balances holding costs and penalty costs, ensuring that production and storage resources are used at their most efficient levels without over-consuming energy or space.
- **Optimized Supply Cycles:** Determining the correct reorder point (R) ensures that demand is met without the need for urgent, carbon-intensive replenishment shipments.
- **Sustainable Consumption Patterns:** Tracking the "proportion of demands not met" allows for a highly responsive supply chain that satisfies consumer needs while maintaining responsible production volumes.